NumPy Warm-up and Maximum-Likelihood Estimation

Machine Learning, Homework 1

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Notebook. `numpy_exercises.ipynb` is a set of NumPy drills that I completed: creating arrays (`zeros`, `arange`, `linspace`, `eye`, random draws), reshaping, slicing a 5×5 matrix to reproduce given sub-matrices, and reductions (sum, standard deviation, column sums).

Written part. Each problem gives a density and N i.i.d. samples. I wrote the likelihood, took its logarithm, set the derivative to zero and checked that the stationary point is a maximum.

Exponential rate. $f(y;\lambda) = \lambda e^{-\lambda y}$ gives $\ell(\lambda) = N \ln \lambda - \lambda \sum_i y_i$, so $\hat{\lambda} = N / \sum_{i=1}^{N} y_i$.

Power density on $(0,1)$**.** $f(y;a) = a y^{a-1}$ gives $\ell(a) = N \ln a + (a-1) \sum_i \ln y_i$, so $\hat{a}_{\mathrm{ML}} = -\big(\frac{1}{N} \sum_i \ln y_i\big)^{-1}$.

Gaussian mean with a Gaussian prior. With $y_n \sim \mathcal{N}(\mu, 1)$ and $\mu \sim \mathcal{N}(0, \sigma_\mu^2)$, completing the square gives the posterior $\mathcal{N}(\mu_N, \sigma_N^2)$ with $\sigma_N^2 = (N + \sigma_\mu^{-2})^{-1}$ and $\mu_N = \sigma_N^2 \sum_n y_n$. The posterior is Gaussian, so the MAP estimate is its mean, $\hat{\mu}_{\mathrm{MAP}} = \sum_n y_n / (N + \sigma_\mu^{-2})$, which shrinks the sample mean towards the prior mean 0.

Weibull-type density, known shape δ**.** $f(x;\beta,\delta) = \beta\delta x^{\delta-1} e^{-\beta x^\delta}$ gives $\hat{\beta} = n / \sum_i x_i^\delta$.

Shifted exponential. $f(x;\tau) = \tau e^{-\tau(x-3)}$, $x > 3$, gives $\hat{\tau} = n / \sum_i (x_i - 3)$. For the sample $3.15, 3.65, 3.10, 3.30$: $\hat{\tau} = 4/1.2 \approx 3.33$.